TCN IKORODU · INITIAL PRODUCT SPEC · V0.1

Every seat, accounted for.

A booking & management platform for church transport — replacing WhatsApp threads, phone calls and paper manifests with capacity-aware booking, payments, marshal operations and a real audit trail.

Status: INITIAL SPEC — open for deliberation. Final PRD follows team review.

CONTENTS

01 — Why this exists	The problem it replaces	p.03–06
02 — The product	Users, journeys, lifecycle	p.07–11
03 — Money & policy	Payments, refunds, bulk booking	p.12–14
04 — Operations	Marshal, Admin, notifications	p.15–17
05 — Foundations	Security, data, scope, open issues	p.18–21

Companion artifacts: BRD · PM Decision Log · interactive prototype (PROTOTYPE/)

THE PROBLEM

One admin's WhatsApp is the entire system.

No capacity truth

Seats are promised in chat. Nobody knows a bus is full until people are standing at it.

Payment ≠ seat

Members pay the admin directly; the link between a payment and a confirmed seat lives in memory.

Paper manifests

Marshals check boarding off printed lists. No-shows and absences are never recorded.

No history

No booking record survives the week. Refund disputes, planning, reporting — all unanswerable.

Source: BRD + Transport Dept. liaison (field-verified 2026-10-07)

FIELD TRUTH — LIAISON REPORT

How it actually works today.

2–4 buses per programme

Semi-fixed routes and landmarks, adjusted by turnout.

Cash still matters

A real subset of members will never pay online. A manual-payment path is required at launch — digital-only excludes them.

A register exists

Department keeps member registers + contact spreadsheets, exportable to CSV — pre-loaded accounts are viable.

Refund norm

No-shows are non-refundable unless cancelled well before departure — matching the refund-before-event rule already decided.

Six operational answers logged from the Transport Department liaison (thread 5259c5d9)

THE PRODUCT'S ONE JOB

No individual's memory should be able to break the system.

The literal job is booking a bus seat. The real job is replacing a process whose only record of truth is one person's chat thread — so that capacity, payment status and boarding are provable facts, not recollections.

Second job: it has to be adopted by a congregation whose only digital habit is WhatsApp. A perfect engine nobody installs doesn't retire the manual process.

```
member → auth → programme → route/pickup →
capacity check → booking → payment* →
confirmation → marshal boarding →
attendance → cancel/refund* → report →
audit

* where applicable
```

USERS & ROLES

Four faces of the system.

Member

Books seats, pays where applicable, holds a scannable ticket. Can act as a **Host** — booking and paying for a group of beneficiaries.

Marshal

Runs boarding at the pickup point: live manifest, check-in, no-show marking. Assigned to buses by Admin — never self-assigned (D-005).

Admin

Church transport dept. staff. Owns programmes, routes, buses, per-bus booking windows (D-013), refunds, reports.

Executive Sponsor

Church leadership. Approves major policy and budget decisions; reads reports; not a daily operator.

Member · Host · Marshal · Admin · Sponsor

CORE JOURNEY — THE HAPPY PATH

From link to seat in under two minutes.

01
Install & sign in
PWA forced install + member gate

02
Pick programme
Sunday free · special paid

03
Route + bus
live seats-left visible

04
Pay or confirm
Paystack · seat held during checkout

05
Ticket issued
booking ref + push confirm

06
Board
marshal check-in → attendance logged

Every step writes an audit event. Every state change is attributable to an actor.

BOOKING LIFECYCLE

A booking is a state, not a note.

draft → held → payment pending → confirmed → boarded

held expired · payment failed · cancelled (refundable window) · no-show · refunded

Held

Seat reserved during checkout — TTL hold, auto-released on expiry. (Mechanism pending OI-001 reconvene; TTL-hold is the engineering recommendation.)

No-show (D-008)

A first-class terminal state — not a footnote. Feeds reporting and future forecasting.

Immutable transitions

Every transition is an append-only audit row: actor, before/after, timestamp, context.

CAPACITY MODEL

Routes carry buses. Buses carry seats.

D-001: a Route has many Buses; route capacity = sum of its buses. **D-013:** Admin opens/closes booking per individual bus, not just per route.

D-004: booking windows are scheduler-driven — open Friday noon, close before departure, per programme.

Route: Ikorodu Garage → Church	2 buses	open
Bus 1 · departs 6:30	13/14 booked	1 left
Bus 2 · departs 7:00	11/14 booked	3 left
Route: Ogolonto → Church	1 bus · no marshal	closed

Invariant: confirmed + held ≤ capacity, enforced inside one transaction — never in application code alone.

ENTS

Free on Sunday. Paid when it's special.

Sunday services — free

Booking confirms instantly; no payment step.

Special programmes — Paystack

Hosted checkout; seat is held while the member pays (held state auto-expires — 15 min proposed).

Manual / cash path

Liaison-verified requirement: cash and pay-the-admin bookings stay possible at launch, recorded against the same manifest.

Webhook-confirmed

Payment truth comes from Paystack webhooks — idempotent, replay-safe, never from the browser's word alone.

BULK PAYMENT — IN MVP (D-011)

One Host pays. Many ride.

A **BookingGroup** lets a Host book and pay for beneficiaries — family, a cell group, elderly members. Beneficiaries appear on the manifest as individuals; the Host carries the payment.

Schema-altering, so it's pulled into MVP rather than retrofitted later.

```
BookingGroup
host: Adaeze O. (payer)
members: [self, +3 beneficiaries]
payment: ₦10,000 · Paystack
seats: 4 × Bus 1 · held→confirmed
manifest: 4 individual rows
```

Open design point raised by Backend: partial cancellation of one beneficiary inside a paid group — on the OI-001 agenda

CANCELLATION & REFUND

Before the event, money back. After it, the seat is spent.

Cancel before event day
Auto-refund via Paystack refund API (D-006) — no admin touch

Cancel after cutoff
Seat released to waitlist; payment retained per refund norm

No-show
Terminal state — non-refundable, recorded (D-008)

Field-verified: today's norm is exactly this — cancellation well before departure is the only refundable path.

MARSHAL OPERATIONS

A manifest that survives a 6 AM car park.

Live manifest

Per-bus passenger list with one-tap check-in, no-show marking, and group tags.

Big targets, high contrast

Designed for gloves, glare and haste — the environment is the spec.

Admin-assigned (D-005)

Marshals can't self-assign or release themselves — reassignment is an Admin action, audited.

Offline question open

Liaison asks whether a printable/offline fallback is needed for dead zones — logged for the reconvene agenda.

Replaces: printed paper lists + manual tick-offs (today's actual process)

ADMIN DASHBOARD

The operations room.

01
Bookings
search, filter, cancel, manual entry

02
Fleet
routes, buses, per-bus open/close

03
Payments
transactions, refund queue, eligibility

04
Marshals
assignments, live boarding counts

05
Reports
utilization, no-shows, revenue

06
Audit
append-only log, attributable actions

Interactive mockup exists: **PROTOTYPE/admin.html —** responsive (sidebar desktop / bottom nav mobile), same design system as this deck.

DISTRIBUTION & NOTIFICATIONS

No app store. The link is the store.

Forced PWA install (D-009)

Sign-up gates on installing the PWA — that's the membership distribution mechanism and the push-notification prerequisite on iOS.

Push + email baseline

Booking confirmations, reminders, boarding alerts. SMS/WhatsApp deferred to Phase 2 (budget-gated).

Designed for WhatsApp natives

Every screen must feel more trustworthy than texting the admin — or nothing gets retired.

SECURITY, IDENTITY & AUDIT

Members-only, provable, attributable.

Membership gate

Pre-load from the department register (CSV — confirmed to exist). Self-registration joins only by matching the roll. Duplicate-handling rule pending (D-007).

RBAC

Member / Host / Marshal / Admin / Sponsor scopes. Marshal sees own manifest only.

Append-only audit

Bookings, payments, check-ins, admin actions — immutable event rows. Entity mutability list pending (OI-003).

Rate limits & quotas

Per-profile usage quotas derived from observed workrate (§3.7) — security + infrastructure protection.

Invariant: validated inputs only at trust boundaries; failures are recorded objects, not silent catches

DATA & ANALYTICS

Utilization first. Prediction when it's earned.

MVP: utilization dashboard

Bookings, confirmations, no-shows, route/bus fill — the Admin's operating numbers (BRD §17).

Phase 2: quota/workrate engine

30-day rolling activity windows feed per-profile rate limits — once real data exists to calibrate them.

Deferred: no-show forecasting

ML becomes a decision tool only after ~6 months of complete history — Data Science's recommendation, adopted.

30d

rolling window chosen for workrate measurement — responsive to real behaviour, not stale averages.

SCOPE — WHAT SHIPS WHEN

Everything has a tier.

CAPABILITY	NOTES	TIER
Booking + capacity + manifests	core loop	MVP
Paystack payments + auto-refund	before-event rule	MVP
Bulk/group booking	BookingGroup	MVP
Manual/cash payment path	field-verified need	MVP
PWA + push + email	forced install	MVP
Audit trail + RBAC	append-only	MVP
Rate-limit quotas	workrate-derived	MVP-lite
SMS/WhatsApp notify	budget-gated	PHASE 2
Managed DB backups	$0 constraint now	PHASE 2
No-show ML forecasting	needs history	PHASE 2
Multi-product tenancy	namespacing now	FUTURE

OPEN FOR DELIBERATION — HONEST STATUS

What's not decided yet.

OI-001 · Technology stack	gates all schema + backend work	BLOCKED
D-003 · Seat-hold mechanism	TTL hold table recommended by Backend	RECONVENE
D-007 · Duplicate registration	resolve with AUTH-2 register pre-load	RECONVENE
Marshal manifest lock + offline	cutoff time, printable fallback — liaison's questions	OPEN

Rule: respondents reconvene — PM does not break ties (D-012).
Everything above carries a named owner and a recommended default so work isn't blind.

NEXT MOVES

Deliberate, then we build.

- **This spec → team deliberation → final PRD**
- **OI-001 reconvene — stack, seat-hold, duplicates**
- **Schema + API contracts release at Gate 0**
- **Prototype → validated flows → build**

Read alongside: BRD · PM Decision Log & Roadmap · PROTOTYPE/ (member · marshal · admin mockups).